Proposition 1: Suppose $v_1, v_2, \dots, v_n$ span a vector space V, and suppose $F: V \to U$ is linear. Then $F(v_1), F(v_2), \dots, F(v_n)$ span $Im\ F$.

Definition:

Let $F: V \to U$ be linear mapping. The rank of F is defined to be the dimension of its image, and the nullity of F is defined to be the dimension of its Kernel: namely,

$rank(F) = \dim(Im\ F)$ and $nullity(F) = \dim\ (Ker\ F)$

Rank-Nullity Theorem:

Let V be of finite dimension, and let $F: V \to U$ be linear. Then

$$\dim V = \dim(Ker\ F) + \dim(Im\ F) = nullity\ (F) + rank\ (F).$$

Problems:

1) Let $F: R^4 \to R^3$ be the linear mapping defined by $F(x, y, z, t) = (x - y + z + t, 2x - 2y + 3z + 4t, 3x - 3y + 4z + 5t)$
   (a) Find a basis and the dimension of the image F.
   (b) Find a basis and the dimension of the Kernel of the map F.

Solution: (a) First find the image of the usual basis vectors of R^4.

$$F(1,0,0,0) = (1,2,3), F(0,1,0,0) = (-1,-2,-3),$$

$$F(0,0,1,0) = (1,3,4), F(0,0,0,1) = (1,4,5)$$

By Proposition 1, the image vectors span $Im\ F$. Hence from the matrix M whose rows are these image vectors and row reduced to echelon form.

$$M = \begin{bmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 1 & 3 & 4 \\ 1 & 4 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 2 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus $(1,2,3)$ and $(0,1,1)$ form a basis of $Im\ F$. Hence $\dim(Im\ F) = 2$ and

$$rank(F) = 2.$$

(b) Set $F(v) = 0$, where $v = (x, y, z, t)$
$F(x, y, z, t) = (x - y + z = t, 2x - 2y + 3z + 4t, 3x - 3y + 4z + 5t) = (0,0,0)$

Set corresponding components equal to form the following homogenous system whose solution space is $Ker\ F$:

$$x - y + z + t = 0, 2x - 2y + 3z + 4t = 0, 3x - 3y + 4z + 5t = 0$$

$$=> x - y + z + t = 0, \qquad z + 2t = 0$$

The free variables are y and t. Hence $\dim(Ker\ F) = 2$ or $nullity(F) = 2$.

(i) Set $y = 1, t = 0$ to obtain the solution $(-1,1,0,0)$,
(ii) Set $y = 0, t = 1$ to obtain the solution $(1,0,2,1)$.

By rank nullity theorem, $\dim(Im\ F) + \dim(\ker F) = 4 = R^4$.

2) Let $F: R^4 \to R^3$ be the linear mapping defined by $F(x, y, z, t) = (x - y + z + t, x + 2z - t, x + y + 3z - 3t)$

Find a basis and the dimension of: (a) the image of F, (b) the kernel of F.

(a) Find the images of the usual basis of R^4:

$$F(1,0,0,0) = (1,1,1), F(0,1,0,0) = (-1,0,1),$$
$$F(0,0,1,0) = (1,2,3), F(0,0,0,1) = (1,-1,-3)$$

By Proposition 1, image vectors span $Im\ F$. Hence from the matrix whose rows are these image vectors, and row reduced to echelon form:

$$\begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \\ 1 & 2 & 3 \\ 1 & -1 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \\ 1 & -2 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus $(1,1,1)$ and $(0,1,2)$ forms a basis for $Im\ F$: hence $\dim(Im\ F) = 2$.

(b) Set $F(v) = 0$, where $v = (x, y, z, t)$; that is, set

$$F(x, y, z, t) = (x - y + z + t, x + 2z - t, x + y + 3z - 3t) = (0,0,0)$$

Set corresponding entries equal to each other to form the following homogenous system whose solution space is $Ker\ F$:

$$x - y + z + t = 0, x + 2z - t = 0, x + y + 3z - 3t = 0$$

$$=> x - y + z + t = 0, y + z - 2t = 0$$

The free variables are z and t. Hence $\dim(Ker\ F) = 2$.

(i) Set $z = -1, t = 0$ to obtain the solution $(2,1,-1,0)$.

(ii) Set $z = 0, t = 1$ to obtain the solution $(1,2,0,1)$.

Thus $(2,1,-1,0)$ and $(1,2,0,1)$ form a basis of $Ker\ F$.

3) Consider the matrix mapping $A: R^4 \to R^3$, where $A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 3 & 5 & -2 \\ 3 & 8 & 13 & -3 \end{bmatrix}$. Find a basis and the dimension of: (a) the image of A, (b) the kernel of A.

Solution:

(a) The column space of A is equal to $Im\ A$. Now reduce A^T to echelon form:

$$A^T = \begin{bmatrix} 1 & 1 & 3 \\ 2 & 3 & 8 \\ 3 & 5 & 13 \\ 1 & -2 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 2 \\ 0 & 2 & 4 \\ 0 & -3 & -6 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus $\{(1,1,3), (0,1,2)\}$ is a basis of $Im\ A$, and $\dim(Im\ A) = 2$.

(b) Here Ker A is the solution space of the homogenous system $AX = 0$, where $X = \{x, y, z, t\}^T$.

Thus reduce the matrix A of coefficient to echelon form:

$$\begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 3 & 5 & -2 \\ 3 & 8 & 13 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & 1 & 2 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Or $x + 2y + 3z + t = 0, y + 2z - 3t = 0$

The free variables are z and t. Thus $\dim(Ker\ A) = 2$.

(i) Set $z = 1, t = 0$ to get the solution $(1,-2,1,0)$.

(ii) Set $z = 0, t = 1$ to get the solution $(-7,3,0,1)$.

Thus $(1,-2,1,0)$ and $(-7,3,0,1)$ form a basis for $Ker\ A$.

Exercise:

1) Let $G: R^3 \to R^3$ be the linear mapping defined by $G(x, y, z) = (x + 2y - z, y + z, x + y - 2z)$.
   Find a basis and the dimension of : (a) the image of G, (b) the kernel of G.
2) Let $F: R^4 \to R^3$ defined by $F(x, y, z, t) = (x + 2y + 3z + 2t, 2x + 4y + 7z + 5t, x + 2y + 6z + 5t)$. Find a basis and the dimension of : (a) the image of F, (b) the kernel of F.
3) Each of the following matrices determines a linear map from R^4 into R^3:
   (a) $A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & -1 & 2 & -1 \\ 1 & -3 & 2 & -2 \end{bmatrix}$, (b) $B = \begin{bmatrix} 1 & 0 & 2 & -1 \\ 2 & 3 & -1 & 1 \\ -2 & 0 & -5 & 3 \end{bmatrix}$
   Find a basis as well as the dimension of the kernel and the image of each linear map.

Matrix representation of a Linear operator:

Let T be a linear operator (transformation) from a vector space V into itself, and suppose $S = \{u_1, u_2, \dots, u_n\}$ is a basis of V. Now $T(u_1), T(u_2), \dots, T(u_n)$ are vectors in V, and so each is a linear combination of the vectors in the basis S; say,

$$\begin{aligned} T(u_1) &= a_{11}u_1 + a_{12}u_2 + \cdots + a_{1n}u_n \\ T(u_2) &= a_{21}u_1 + a_{22}u_2 + \cdots + a_{2n}u_n \\ &\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots \\ T(u_n) &= a_{n1}u_1 + a_{n2}u_2 + \cdots + a_{nn}u_n \end{aligned}$$

The following definition applies.

Definition: The transpose of the above matrix of coefficients, denoted by $m_S(T)$ or $[T]_S$, is called the matrix representation of T relative to the basis S, or simply the matrix T in the basis S.

Using the coordinate (column) vector notation, the matrix representation of T may be written in the form $m_S(T) = [T]_S = [[T(u_1)_S, [T(u_2)]_S, \dots, [T(u_n)]_S]$.

That is, the columns of $m(T)$ are the coordinate vectors of $T(u_1), T(u_2), \dots, T(u_n)$, respectively.

Example 1: Let $F: R^2 \to R^2$ be the linear operator defined by $F(x, y) = (2x + 3y, 4x - 5y)$.

(a) Find the matrix representation of F relative to the basis $S = \{u_1, u_2\} = \{(1,2), (2,5)\}$.
Solution: First find $F(u_1)$ and then write it as linear combination of the basis vectors u_1 and u_2.
We have $F(u_1) = F\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 8 \\ -6 \end{bmatrix} = x\begin{bmatrix} 1 \\ 2 \end{bmatrix} + y\begin{bmatrix} 2 \\ 5 \end{bmatrix}$

Implies $x + 2y = 8, 2x + 5y = -6$.

Solving these, we get $x = 52, y = -22$. Hence $F(u_1) = 52u_1 - 22u_2$.

Next find $F(u_2)$, and then write it as a linear combination of u_1 and u_2.

$$F(u_2) = F\left(\begin{bmatrix}2\\5\end{bmatrix}\right) = \begin{bmatrix}19\\-17\end{bmatrix} = x\begin{bmatrix}1\\2\end{bmatrix} + y\begin{bmatrix}2\\5\end{bmatrix}$$

Implies $x + 2y = 19, 2x + 5y = -17$.

Solving these, we obtain $x = 129, y = -55$. Hence $F(u_2) = 129u_1 - 55u_2$.

Now write the coordinates of $F(u_1)$ and $F(u_2)$ as columns to obtain the matrix

$$[F]_S = \begin{bmatrix}52 & 129\\-22 & -55\end{bmatrix}$$

(b) Find the matrix representation of F relative to the basis $E = \{e_1, e_2\} = \{(1,0), (0,1)\}$.
Solution: $F(e_1) = F(1,0) = (2, 2) = 2e_1 + 4e_2$
$F(e_2) = F(0,1) = (3, -5) = 3e_1 - 5e_2$
And so $[F]_E = \begin{bmatrix}2 & 3\\4 & -5\end{bmatrix}$.
Note that the coordinates of $F(e_1)$ and $F(e_2)$ form the columns, not the rows, of $[F]_E$. Also, note that the arithmetic is much simpler using the usual basis of R^2.

Example 2: Let V be the vector space of functions with basis $S = \{\sin t, \cos t, e^{3t}\}$, and let $D: V \to V$ be the differential operator defined by $D\big(f(t)\big) = \frac{d(f(t))}{dt}$.
We compute the matrix representing D in the basis S:

$$D(\sin t)_ = \cos t = 0(\sin t) + 1\ (\cos t) + 0(e^3 t)$$

$$D(\cos t) = -\sin t = -1(\sin t) + 0(\cos t) + 0(e^{3t})$$

$$D(e^{3t}) = 3e^{3t} = 0(\sin t) + 0(cost\ ) + 3(e^{3t})$$

And so $[D] = \begin{bmatrix}0 & -1 & 0\\1 & 0 & 0\\0 & 0 & 3\end{bmatrix}$
Note that the coordinates of $D(sin\ t), D(cos\ t), D(e^{3t}\ )$ form the columns, not the rows of $[D]$.

Exercise:

1) Consider the linear mapping $F: R^2 \to R^2$ defined by $F(x, y) = (3x + 4y, 2x - 5y)$ and the following basis of R^2: $E = \{e_1, e_2\} = \{(1,0), (0,1)\}$ and $S = \{u_1, u_2\} = \{(1,2), (2,3)\}$
   (a) Find the matrix A representing F relative to the basis E.
   (b) Find the matrix B representing F relative to the basis S.
2) Let $A: R^2 \to R^2$ be defined by the matrix $A = \begin{bmatrix}5 & -1\\2 & 4\end{bmatrix}$.
   (a) Find the matrix B representing A relative to the basis $S = \{u_1, u_2\} = \{(1,3), (2,8)\}$.
   (b) For $v = (a, b)$, find $[v]_S$ and $[A(v)]_S$.

Change of Basis

Definition: Let $S = \{u_1, u_2, \dots, u_n\}$ be a basis of a vector space V and let $S' = \{v_1, v_2, \dots, v_n\}$ be another basis. Each vector in the new basis S' can be written uniquely as a linear combination of the vectors in S: say,

$$v_1 = a_{11}u_1 + a_{12}u_2 + \cdots + a_{1n}u_n$$

$$v_2 = a_{21}u_1 + a_{22}u_2 + \cdots + a_{2n}u_n$$

$$\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots$$

$$v_n = a_{n1}u_1 + a_{n2}u_2 + \cdots + a_{nn}u_n$$

Let P be the transpose of the above matrix of coefficients: that is, let $P = [p_{ij}]$, where $p_{ij} = a_{ij}$. Then P is called the change of basis matrix (or transition matrix) from the old basis S to new basis S'.

Remark 1: The above change of basis matrix P may also be viewed as the matrix whose column are, respectively, the coordinate column vectors of the new basis vectors v_i relative to the old basis S; namely,

$$P = [[v_1]_S, [v_2]_S, \dots, [v_n]_S]$$

Remark 2: Analogously, there is a change of basis matrix Q from the new basis S' to the old basis S. Similarly, Q may be viewed as the matrix whose columns are, respectively, the coordinate column vectors of the old basis vector u_i relative to the new basis S; namely,

$$Q = [[u_1]_{S'}, [u_2]_{S'}, \dots, [u_n]_{S'}]$$

Remark 3: Since the vectors $v_1, v_2, \dots, v_n$ in the new basis S' are linearly independent, the matrix P is invertible. Similarly, Q is invertible.

Proposition: Let P and Q be the above change of basis matrices. Then $Q = P^{-1}$.

Example 1: Consider the following two basis of R^2: $S = \{u_1, u_2\} = \{(1,2), (3,5)\}$ and $S' = \{v_1, v_2\} = \{(1,-1), (1,-2)\}$

(a) Find the change of basis matrix P from S to the new basis S'.
Solution: Write each of the new basis vectors of S' as a linear combination of the original basis vectors u_1 and u_2 of S.
We have $\begin{bmatrix} 1 \\ -1 \end{bmatrix} = x\begin{bmatrix} 1 \\ 2 \end{bmatrix} + y\begin{bmatrix} 3 \\ 5 \end{bmatrix}$
Implies $x + 3y = 1, 2x + 5y = -1$ which yields $x = -8, y = 3$.
And $\begin{bmatrix} 1 \\ -2 \end{bmatrix} = x\begin{bmatrix} 1 \\ 2 \end{bmatrix} + y\begin{bmatrix} 3 \\ 5 \end{bmatrix}$
Implies $x + 3y = 1, 2x + 5y = -2$ which yields $x = -11, y = 4$.
Thus $v_1 = -8u_1 + 3u_2$ and $v_2 = -11u_1 + 4u_2$.

Hence $P = \begin{bmatrix} -8 & -11 \\ 3 & 4 \end{bmatrix}$.

(b) Find the change of basis matrix Q from the new basis S' back to the old basis S.
Solution: Here we write each of the old basis vectors u_1 and u_2 of S as a linear combination of the new basis vectors v_1 and v_2 of S' which yields
$u_1 = 4v_1 - 3v_2$ and $u_2 = 11v_1 - 8v_2$.
Hence $Q = \begin{bmatrix} 4 & 11 \\ -3 & -8 \end{bmatrix}$.

Example 2: Consider the following basis of R^3:

$E = \{e_1, e_2, e_3\} = \{(1,0,0), (0,1,0), (0,0,1)\}$ and

$S = \{u_1, u_2, u_3\} = \{(1,0,1), (2,1,2), (1,2,2)\}$

(a) Find the change of basis matrix P from the basis E to the basis S.
Solution: Since E is the usual basis, we can immediately write each basis element S as a linear combination of the basis elements of E. Specifically,

$$u_1 = (1,0,1) = e_1 + e_3, u_2 = (2,1,2) = 2e_1 + e_2 + 2e_3,$$
$$u_3 = (1,2,2) = e_1 + 2e_2 + 2e_3$$

Hence $P = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 1 & 2 & 2 \end{bmatrix}$.
Note: P is simply the matrix whose columns are the basis vectors of S. This is true only because the original basis was the usual basis E.

(b) Find the change of basis matrix Q from the basis S to the basis E.
Solution: Write each of the new basis vectors of E as a linear combination of the original basis elements of S. This yields

$$e_1 = (1,0,0) = -2u_1 + 2u_2 - u_3, e_2 = (0,1,0) = -2u_1 + u_2,$$
$e_3 = (0,0,1) = 3u_1 - 2u_2 + u_3$.
And hence $Q = \begin{bmatrix} -2 & -2 & 3 \\ 2 & 1 & -2 \\ -1 & 0 & 1 \end{bmatrix}$.

Exercise:

1) Consider the following basis for R^2:
$E = \{e_1, e_2\} = \{(1,0), (0,1)\}$ and $S = \{u_1, u_2\} = \{(1,3),\ (1,4)\}$

(a) Find the change of basis matrix P from the usual basis E to S.
(b) Find the change of basis matrix Q from the usual basis E to S.

2) The vectors $u_1 = (1,2,0), u_2 = (1,3,2), u_3 = (0,1,3)$ form a basis S of R^3.
Find:
(a) The change of basis matrix P from the usual basis $E = \{e_1, e_2, e_3\}$ to S.
(b) The change of basis matrix Q from S back to E.